Theo Bernardin

LEAD DATA ENGINEER

linkedin.com/in/theo-bernardin | France (Open to Relocation)

PROFESSIONAL SUMMARY

Lead Data Engineer with over 4 years of experience designing and operating scalable data pipelines on GCP, across a production perimeter exceeding 100 billion rows and 100+ TB at up to 99.6% availability. Expert in SQL, Python, BigQuery, and DBT, with deep experience turning complex multi-source supply-chain and retail data into reliable, cost-efficient single sources of truth. Scrum Master for a 10-person squad; mentored and onboarded 5 engineers. Curious, reliable, and solution-oriented, with a strong background in cloud-native data architecture and a proven track record of delivering business value.

TECHNICAL SKILLS

Languages:	Python, SQL, JavaScript, TypeScript, Shell/Bash
Data & Cloud Stack:	GCP (BigQuery, Pub/Sub, cloud function), DBT (Core), Airflow, Redis, Docker, Terraform
Engineering Practices:	Data Architecture, ETL/ELT Design, Data Modeling, Master Data Management (MDM), Data Governance
Workflow:	GitLab (CI/CD), LLM Integration, Agile/Scrum

PROFESSIONAL EXPERIENCE

Lead Data Engineer | Carrefour — Apr 2023 - Present

- **End-to-End Data Ownership:** Own the data engineering for Carrefour France's entire Offer & Supply Chain perimeter, designing, deploying, and operating dozens of ETL/ELT pipelines on GCP/BigQuery (DBT Core) over 100B+ rows and 100+ TB at up to 99.6% availability.
- **Flagship Pipeline - Single Source of Truth:** Architected a Slowly Changing Dimension (SCD Type 2) pipeline consolidating 13 disparate source systems into one historically-accurate model of product assortment across hypermarkets, supermarkets, proximity stores, and warehouses, replacing a fragmented manual process with an audited single source of truth.
- **Cost Optimization:** Engineered custom delta-detection and early-exit logic plus heavy SQL tuning across pipelines, cutting BigQuery compute cost by ~20% (~€100/day, ~€36K/year) while sustaining 20-minute refresh cadences.
- **Reliability & Monitoring:** Built automated fallback mechanisms and real-time alerting that prevent data loss and duplication across failed runs, keeping critical store and warehouse operations continuously supplied with fresh data.
- **Tech Lead & Scrum Master:** Set the technical vision and engineering standards, run code reviews, mentor and onboard 5 engineers (3 as direct reports), and lead Agile ceremonies for a 10-person squad.

  Stack: SQL, DBT Core, Python, GCP, Airflow, GitLab, Jira, Confluence

Independent Data Instructor | Self-employed — Apr 2024 - Present

- **Trained 100+ professionals** upskilling or transitioning into the data field through continuing professional education programs.
- Instruct core modules covering foundational algorithms (logic and problem-solving), scientific computing (data manipulation, numerical analysis), data visualization, and data engineering pipelines/architecture.

Data Engineer | OPCODES — Mar 2022 - Apr 2023

- **Pipeline Reverse-Engineering:** Reverse-engineered existing data pipelines and ingestion scripts in Python to extract relevant transaction data from blockchain logs.
- **Cloud Storage:** Structured and stored large volumes of transaction data efficiently into Google BigQuery database tables.
- **Data Transformation & Cleaning:** Performed data cleaning, transformation, and aggregation to prepare transaction insights for downstream consumption.
- **Dashboard Components:** Built and refined dynamic dashboard components to explore transaction patterns and user behaviors.

  Technologies used: Python, SQL, BigQuery, Docker, JavaScript, TypeScript

Research Data Scientist Intern | UTAD (Portugal) — May 2021 - Jul 2021

- Conducted research on the early detection of plant grapevine leaf diseases using hyperspectral imaging data.
- Leveraged machine learning techniques (Scikit-learn, PCA) to identify early symptoms and reduce data dimensionality for more efficient processing and storage.
- **Research Publication:** Co-authored and published research paper: 'Automatic detection of Flavescence dorée grapevine disease in hyperspectral images using machine learning' in Elsevier's *Procedia Computer Science* (Link: sciencedirect.com/science/article/pii/S1877050921022201).

EDUCATION

M.S. in Engineering (Diplôme d'ingénieur) - Computer Science & Data — 2019 - 2022
Polytech Annecy-Chambéry

Exchange Program in Computer Science — 2021 - 2022
Babes-Bolyai University

Associate Degree in Computer Science (DUT Informatique) — 2017 - 2019
Université de Lorraine

LANGUAGES & HOBBIES

French (Native) | English (Full Professional)

Hobbies: Horology, Karate, Travel